Austere Medical and Disaster Survival Field Reference Guide: Tactical Trauma, Prolonged Care, and Resource-Constrained Protocols

1. MSMAID Airway Pre-Procedure Checklist and Equipment Tiering

In austere environments and Prolonged Field Care (PFC) scenarios, airway compromise represents the second leading cause of potentially survivable battlefield mortality following massive hemorrhage. Complete airway occlusion leads rapidly to asphyxiation and fatal tissue hypoxia within minutes. Managing advanced airways under resource-constrained conditions introduces severe operational friction, including limited supplemental oxygen reserves, restricted access to rapid sequence intubation (RSI) inductors and neuromuscular blocking agents, and variable operator skill sustainment. Implementing a standardized, structured pre-procedure verification tool—specifically the MSMAID framework—is a critical operational mandate. Systematic pre-procedure preparation directly mitigates procedural failure, prevents unmanaged rapid desaturation, minimizes esophageal misplacement, and significantly reduces tactical airway mortality during extended field care.

1.1 MSMAID Framework Analysis

The MSMAID acronym (Machine, Suction, Monitor, Airway, Intravenous access, Drugs) structures the operator's preparation prior to executing invasive airway interventions. Operational readiness is organized across Minimum, Better, and Best capability tiers:

```
[MSMAID PRE-PROCEDURE VERIFICATION FRAMEWORK]
│
├── M - MACHINE ──────► Min: BVM + PEEP (5 cm H₂O) | Better: Auto Vent | Best: Full-Feature Vent + O₂
├── S - SUCTION ──────► Min: Syringe + NPA | Better: Manual Bulb | Best: Powered Suction + In-Line Adapter
├── M - MONITOR ──────► Min: SpO₂ + PFC Sheet | Better: Capnometer (ETCO₂) | Best: Auto Vitals + Waveform Capnography
├── A - AIRWAY ──────► Min: Ketamine Cric Prep | Better: + Continuous Sedation | Best: RSI + Bougie + Sedation
├── I - IV/IO ACCESS ─► Min: IM/IN Contingency | Better: 1–2x Patent Lines | Best: 2–3x Patent Lines + Standby IO
└── D - DRUGS ────────► Min: Ketamine IM | Better: IV/IO Sedation Pushes | Best: Ketamine Induction/Drip + Opioid Push
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Machine (Post-Intubation Ventilation Equipment)

- **Minimum:** Bag-Valve-Mask (BVM) equipped with a Positive End-Expiratory Pressure (PEEP) valve set to an initial baseline of $5\text{ cm H}_2\text{O}$. When an invasive airway bypasses the glottis, natural physiological PEEP is eliminated; applying mechanical PEEP prevents end-expiratory alveolar collapse and atelectasis. Manual ventilation must deliver controlled, synchronous breaths (no more than 50% of an adult bag volume, or sufficient to initiate visible chest rise) at a rate of 12–16 breaths per minute, avoiding hyperventilation.
- **Better:** Automated portable ventilator (preferably with integrated PEEP controls) paired with a portable oxygen concentrator.

- **Best:** Full-feature portable mechanical ventilator capable of multiple ventilatory modes and precise PEEP titrations, augmented by supplemental oxygen cylinders. Telemedicine consultation should be established when configuring unfamiliar mechanical ventilators.

Suction (Secretions and Airway Clearance)

- **Minimum:** Improvised field suction (e.g., a large-bore syringe fitted to a modified Nasopharyngeal Airway NPA) combined with lateral recovery positioning, provided spinal immobilization constraints do not dictate otherwise.
- **Better:** Manual suction bulb system with specialized airway adapters.
- **Best:** Powered commercial suction apparatus equipped with an oral rigid tip (Yankauer) and an in-line endotracheal tube (ETT) suction adapter.
- *Clinical Directive:* Direct suctioning is vital to clear blood, emesis, and mucus during initial laryngoscopy or cricothyroidotomy. For intubated casualties experiencing elevated peak airway pressures or suspected mucus plugging, perform in-line suctioning. If secretions are thick, instill a $1\text{--}2\text{ mL}$ sterile saline flush prior to catheter insertion. *Suction pressure must only be applied during the controlled withdrawal of the catheter* , never during initial insertion into the airway lumen.

Monitor (Physiological Surveillance and Telemedicine)

- **Minimum:** Pulse oximetry (SpO_2), manual vital signs recorded on a standardized PFC flow sheet by a designated assistant, and voice/data connections for periodic telemedicine reporting.
- **Better:** Portable capnometry (ETCO_2) in addition to continuous SpO_2 , complemented by smartphone digital image transmission for remote physician consultation.
- **Best:** Automated multiparameter vital signs monitor displaying continuous SpO_2 , real-time waveform capnography, electrocardiogram (EKG), and synchronous, live video telemedicine streaming.
- *TBI Target:* For traumatic brain injury (TBI) casualties exhibiting signs of acute cerebral herniation, hyperventilation must be strictly titrated using waveform capnography to maintain an ETCO_2 target between $30\text{ and }35\text{ mmHg}$.

Airway (Primary and Rescue Adjuncts)

- **Minimum:** Operator prepared and equipped for immediate awake surgical cricothyroidotomy using procedural ketamine analgesia.
- **Better:** Ability to perform cricothyroidotomy or basic airway placement paired with continuous post-procedure sedation protocols.
- **Best:** Full Rapid Sequence Intubation (RSI) capability incorporating neuromuscular blockade (chemical paralysis), advanced suction, and continuous intravenous sedation.
- *Bougie Utility:* The gum elastic bougie (Eschmann stylet) is a primary tactile feedback tool. Upon insertion into the trachea, tactile discrimination allows the operator to feel the tip click against the tracheal rings or encounter a definitive tactile "stop" at the carina (typically at $24\text{--}40\text{ cm}$). The ETT is then

advanced smoothly over the bougie into the trachea, minimizing blind insertion failure. The bougie also facilitates rapid tube replacement during mechanical tube displacement or cuff failure.

Intravenous Access and Drugs

- **Access Tiers:**
- *Minimum:* Intramuscular (IM) or intranasal (IN) route contingencies for immediate procedural sedation if intravenous (IV) or intraosseous (IO) lines fail or cannot be rapidly established. IV/IO placement attempts continue once the airway is secured.
- *Better:* $1\text{--}2$ patent, large-bore IV or IO lines.
- *Best:* $2\text{--}3$ patent IV/IO lines with a dedicated secondary IO device held on standby.
- **Drug Preparation:**
- *Induction:* Procedural-dose ketamine ($1\text{--}2\text{ mg/kg}$ IV push) combined with local subcutaneous infiltration of $1\text{--}2\%$ lidocaine over the cricothyroid membrane for surgical airways.
- *Prolonged Sedation:* Minimum post-airway sedation requires ketamine ($3\text{--}4\text{ mg/kg}$ IM). Advanced tiers utilize titrated IV/IO ketamine drips, augmented by hydromorphone or alternate opioids for breakthrough pain and midazolam for amnestic sedation.

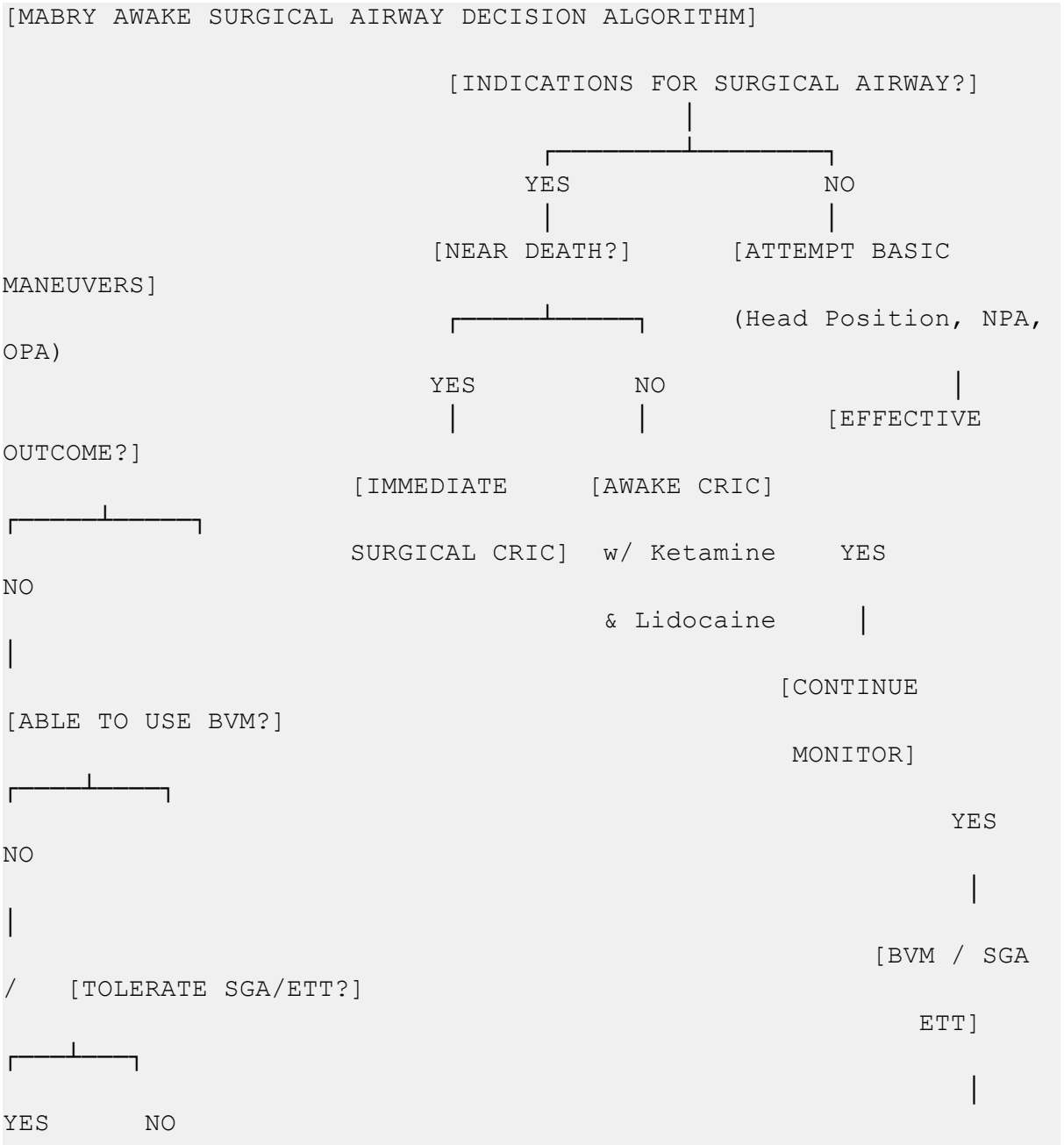
1.2 Airway Adjunct Assessment Table

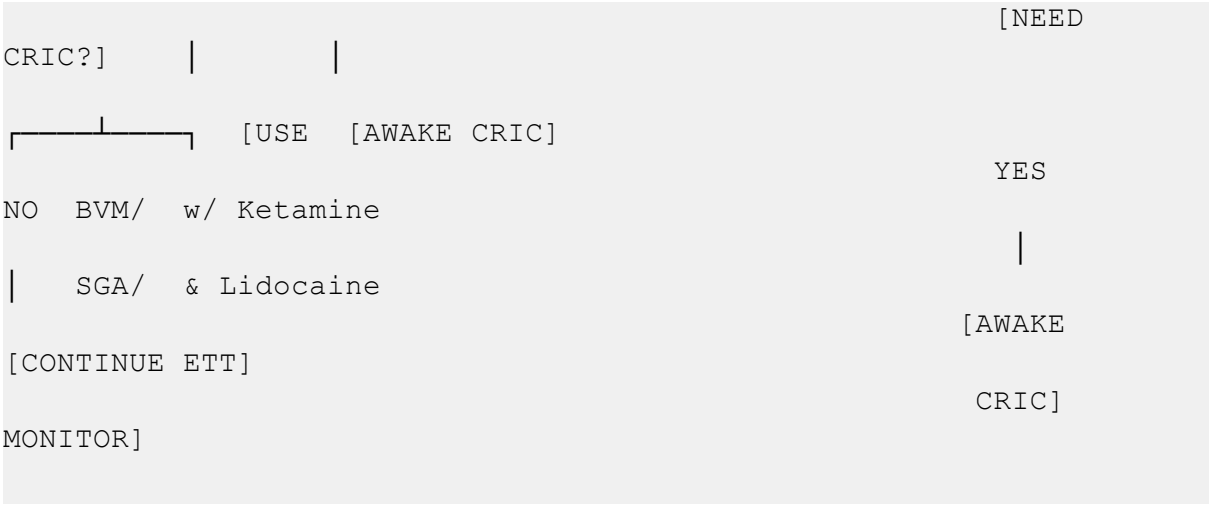
Device / Technique,Pros,Cons,Pharmacological Requirements to Maintain ($0 / + / ++$),Required Skill Level
Head-Tilt / Chin-Lift,Non-invasive; requires no equipment; immediate execution.,Temporary; does not protect against aspiration or soft tissue obstruction; hands-intensive.,0 (No additional medications required),Non-Medic & Medic
Nasopharyngeal Airway (NPA),Well tolerated in semi-conscious casualties; simple insertion; preserves spontaneous drive.,Does not prevent aspiration; potential for nasal mucosa bleeding; easily clogged by thick secretions.,0 (No additional medications required),Non-Medic & Medic
Supraglottic Airway (SGA),Faster placement than ETT; does not require direct laryngoscopy or surgical incisions.,Noxious stimulus in conscious casualties; elevated risk of gastric inflation and aspiration over extended durations.,$+$ (Intermittent dosing or mild sedation required),Medic
Oral Endotracheal Tube (ETT),Definitive airway with inflated tracheal cuff; familiar to advanced clinicians; direct visualization.,High skill decay; requires neuromuscular blockade (RSI) and direct visualization; high failure rate in severe facial trauma.,$++$ (Continuous sedation and potential paralysis required),M+ (Medic with advanced training / specialized orotracheal intubation sustainment)
Surgical Cricothyroidotomy,Definitive surgical airway; bypasses severe upper anatomical or facial disruption; well tolerated; requires minimal sedation once secured.,Invasive surgical procedure; high failure rates under stress without structured training; hemorrhage risk.,$+$ (Intermittent dosing or localized infiltration required),Medic
Note for Pharmacological Requirements: $0 = \text{no additional medications}$; $+ = \text{intermittent or procedural medications}$; $++ = \text{continuous sedation or multiple agents required}$ .Strict adherence to structured pre-procedure checklists directly

establishes control over the casualty's oxygenation and ventilation, creating a stable foundation for managing complex anatomical or maxillofacial trauma.

2. Clinical Indications, Contraindications, and Execution of Surgical Cricothyroidotomy

Surgical cricothyroidotomy represents the definitive surgical pathway for establishing an invasive airway in tactical trauma management. In civilian prehospital settings, advanced airways are predominantly placed in elderly patients experiencing non-traumatic cardiac arrest; conversely, military combat casualties requiring emergency field airways present almost exclusively with severe traumatic injury to the head, face, or neck. Distorted tissue planes, active hemorrhage, and unsterile operational conditions render oral endotracheal intubation exceptionally difficult and prone to failure. When direct laryngoscopy cannot be performed or maintained safely, surgical cricothyroidotomy serves as the primary definitive airway intervention to prevent asphyxiation.





2.1 Anatomical Disruption and Clinical Indications

An invasive airway must be established promptly when casualties present with severe structural disruption or critical respiratory failure.

Disrupted Face or Neck Anatomy

- **Massive Maxillofacial Trauma:** Fractures and soft-tissue avulsions causing pharyngeal structural collapse, glossal displacement, or uncontrolled hemorrhage that occludes the upper airway.
- **Thermal Burns and Inhalation Injury:** Facial burns, soot deposition, or steam inhalation presenting with progressive hoarseness, stridor, or supraglottic edema threatening rapid closure.
- **Massive Penetrating or Blunt Neck Trauma:** Direct mechanical disruption of the larynx, hypopharynx, or cervical trachea destroying standard visual landmarks.
- **Expanding Neck Hematomas:** Rapidly accumulating blood beneath deep cervical fascia causing tracheal deviation and extraluminal airway compression.
- **Acute Pharyngeal Infections:** Deep space swelling (e.g., retropharyngeal abscess, epiglottitis) physically obstructing upper airflow.
- **Uncleared Foreign Body Obstruction:** Mechanical upper airway impaction refractory to standard non-invasive clearing maneuvers.
- **Severe Anaphylaxis:** Progressive, rapid-onset angioedema of the tongue, soft palate, and larynx unresponsive to epinephrine.

Insufficient Oxygenation or Ventilation

- **Severe Chest Wall and Pulmonary Trauma:** Flail segments, massive blast contusions, or severe pulmonary lacerations requiring prolonged positive-pressure ventilation.
- **Extensive Thermal Burns ($>40\%$ TBSA):** Massive systemic fluid resuscitation driving progressive systemic and airway tissue edema.
- **Traumatic Brain Injury ($\text{GCS} \le 8$):** Severe central nervous system depression causing loss of protective airway reflexes and hypoventilation.
- **TBI with Imminent Cerebral Herniation:** Acute intracranial hypertension requiring temporary hyperventilation strictly titrated to maintain an ETCO_2 between $30\text{ and }35\text{ mmHg}$.

- **Requirement for Deep Procedural Sedation:** Long-duration surgical interventions or invasive procedures in field care requiring respiratory support.
- **Acute Severe Respiratory Failure:** Pulmonary pathology including chemical or toxic gas inhalation, ARDS secondary to massive transfusion or sepsis, severe pneumonia, massive pulmonary embolism, TRALI, or TACO.

2.2 Decision Matrix and Execution Protocol

The Mabry Awake Surgical Airway Algorithm governs clinical decision-making in the field:

```
[MULTIMODAL AIRWAY VERIFICATION PROTOCOL]
│
├── 1. DIRECT VISUALIZATION ──► Observe tube tip passing between vocal cords (ETT)
├── 2. AUSCULTATION ──────────► Epigastrium: SILENT | Left/Right Axillae: EQUAL BREATH SOUNDS
├── 3. CAPNOGRAPHY ───────────► Colorimetric Change + Waveform Capnography (Target: 35-45 mmHg)
├── 4. PHYSICAL MARKS ────────► Equal Bilateral Chest Rise + Tube Misting + No Gastric Distension
└── 5. ULTRASOUND / EDD ──────► Direct Tracheal Visualization + Endotracheal Detection Device
```

Algorithmic Pathway

- **Initial Assessment:** Evaluate for primary indications for a surgical airway.
- **Near-Death Status:** If surgical indications exist and the casualty is in extremis (near-death status), execute an **Immediate Surgical Cricothyroidotomy** without delay for pharmacological induction.
- **Awake Surgical Airway Pathway:** If the casualty is conscious or semi-conscious and not in immediate cardiac arrest, perform an **Awake Surgical Cricothyroidotomy** under local anesthesia ($1\text{--}2\%$ lidocaine infiltration) combined with procedural ketamine ($1\text{--}2\text{ mg/kg}$ IV push or $3\text{--}4\text{ mg/kg}$ IM).
- **Non-Surgical Contingency:** If surgical indications are absent, attempt non-invasive stabilization (head repositioning, chin-lift, NPA).
- If non-invasive measures fail, evaluate BVM ventilation efficacy.
- If BVM is successful, maintain ventilation or consider SGA/ETT placement. If the casualty deteriorates or develops secondary airway compromise, transition directly to awake surgical cricothyroidotomy.
- If the casualty cannot be ventilated via BVM and cannot tolerate SGA or ETT placement, perform an immediate surgical cricothyroidotomy.
- *Telemedicine Directive:* In urgent but non-emergent scenarios, establish a telemedicine link to verify decision-making prior to making surgical incisions.

Multi-Modal Tube Verification Protocol

Misplacement of an airway tube into the hypopharynx, esophagus, or right mainstem bronchus is rapidly fatal if uncorrected. Confirm placement using all available metrics:

- **Direct Visualization:** Observe the ETT cuff pass completely beyond the vocal cords.

- **Auscultation:** Auscultate the epigastrium (must be completely silent, with no gastric gurgling) followed by equal bilateral auscultation at the axillary lines and pulmonary apices.
- **Capnography:** Confirm colorimetric change (yellow) or verify continuous waveform capnography (ETCO_2). Capnography is the definitive standard to rule out esophageal intubation.
- **Physical Indicators:** Confirm symmetrical bilateral chest rise and fall, observe condensation (misting) inside the clear lumen of the tube, and confirm the complete absence of progressive abdominal/gastric distension.
- **Auxiliary Verification:** Utilize an Endotracheal Detection Device (EDD) or point-of-care ultrasound (POCUS) to confirm intra-tracheal position relative to adjacent vascular structures. Reassess verification metrics frequently, particularly after casualty movements.Securing a definitive airway stabilizes central pulmonary gas exchange, allowing clinical focus to shift toward systemic vascular management and hemorrhage control protocols.

3. Tourniquet Management: Conversion Protocols, Golden Windows, and Dead Limb Dynamics

Extremity hemorrhage represents the primary cause of preventable death on the battlefield. While emergency application of a high-and-tight arterial limb tourniquet is a life-saving intervention, maintaining prolonged arterial occlusion introduces severe downstream complications. Extended limb ischemia drives tissue necrosis, nerve compression, systemic compartment release, and metabolic derangements. In Prolonged Field Care, the operator must actively manage tourniquet conversion to balance life-threatening hemorrhage control against the risks of limb loss and fatal reperfusion shock.

3.1 Tourniquet Conversion Golden Windows

```
[TOURNIQUET TIME WINDOWS & CLINICAL DIRECTIVES]
│
├── < 2 HOURS ──────────► SAFE CONVERSION WINDOW
│                         High success rate; minimal ischemic risk.
Convert to pressure/hemostatic dressing.
│
├── 2 TO 6 HOURS ───────► CAUTIONARY CONVERSION WINDOW
│                         Progressive ischemic injury. Convert ONLY
under direct monitoring & fluid readiness.
│
└── > 6 HOURS ──────────► ABSOLUTE CONTRAINDICATION ("DEAD LIMB
RULE")
                          Extensive tissue necrosis. DO NOT REMOVE
TOURNIQUET.
                          Release causes fatal reperfusion shock,
hyperkalemic arrest, & massive acidosis.
```

Critical Timing Thresholds

- **Ischemia Under 2 Hours:** The ideal window for tourniquet conversion. Tissue damage is minimal, and conversion to a pressure dressing or hemostatic bandage carries a high success rate without severe metabolic penalties.
- **Ischemia Between 2 and 6 Hours:** Intermediate risk window. Progressive skeletal muscle ischemia occurs. Tourniquet conversion should be attempted cautiously if hemodynamics are stable, direct wound visualization is possible, and active hemorrhage can be controlled.
- **Ischemia Exceeding 6 Hours:** Absolute threshold for the **"Dead Limb Rule."** Prolonged arterial occlusion exceeding 6 hours results in irreversible muscular and neural necrosis.

The "Dead Limb Rule" and Reperfusion Dynamics

When a limb has suffered complete arterial occlusion for longer than 6 hours, it becomes cold, mottled, insensate, rigid (rigor-like state), and pulseless. *Under no circumstances should a tourniquet be released from a dead or irreversibly ischemic limb in an austere environment.*Releasing a tourniquet after prolonged ischemia dumps toxic metabolic byproducts into systemic circulation (reperfusion syndrome). Severe cellular lysis within the ischemic muscle releases massive quantities of potassium (K^+), myoglobin, purines, and lactic acid. Systemic release upon tourniquet removal causes sudden, catastrophic reperfusion shock, lethal hyperkalemic cardiac arrest, and acute oliguric renal failure secondary to renal tubular myoglobin precipitation. If an ischemic limb cannot be safely converted within 6 hours, the tourniquet must remain in place until definitive surgical amputation can be performed at a higher echelon of care.

3.2 Tourniquet Conversion Decision Framework

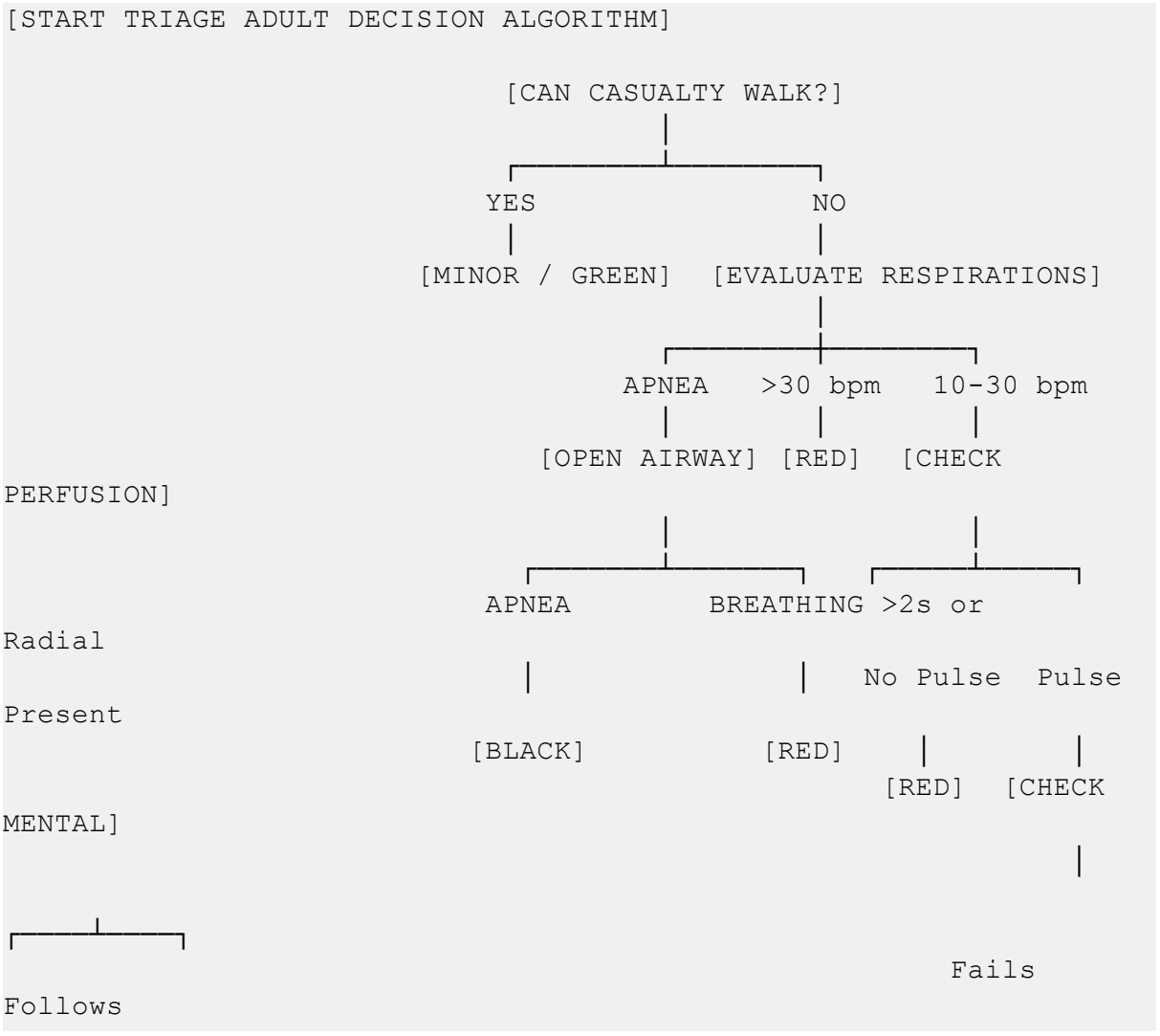
Execute tourniquet conversion systematically using the following protocol:

- **Step 1: Casualty Assessment and Resuscitative Stabilization**
- Verify that the casualty is not in refractory hemorrhagic shock.
- Confirm adequate systemic resuscitation; do not attempt conversion if the casualty demonstrates persistent hypotension ($\text{SBP} < 90\text{ mmHg}$) or uncorrected hypovolemia.
- Ensure adequate pain control and sedation are active prior to wound manipulation.
- **Step 2: Wound Re-Evaluation and Hemostatic Preparation**
- Fully expose the injury site by cutting away clothing to identify all bleeding sources.
- Examine the wound for traumatic amputation. *Do not convert tourniquets on complete or near-complete traumatic amputations.*
- Apply a modern hemostatic gauze dressing directly into the wound cavity, packing tightly against the point of vascular disruption. Secure with a high-pressure dressing.
- **Step 3: Tactical and Operational Reassessment**
- Verify that tactical conditions allow continuous monitoring of the casualty for at least 30–60 minutes following conversion.
- Do not attempt conversion if the tactical scenario prevents immediate detection or management of recurrent massive hemorrhage.
- **Step 4: Incremental Tourniquet Conversion Execution**
- Apply a secondary tourniquet $2\text{--}3\text{ inches}$ above the wound site if the initial tourniquet was applied high-and-tight over clothing.

- Slowly loosen the high-and-tight tourniquet over $1\text{--}2\text{ minutes}$, monitoring the wound site continuously for arterial re-bleeding.
- Leave the loosened tourniquet *in situ* directly on the skin proximate to the wound; if overt arterial hemorrhage recurs and cannot be controlled by the pressure dressing, instantly re-tighten the tourniquet.
- **Step 5: Post-Conversion Reassessment and Documentation**
- Evaluate distal capillary refill, neurological function, and dressing saturation continuously.
- Document the precise time of conversion and mark the casualty's forehead and triage card clearly.Systemic casualty stabilization requires categorizing multiple casualties efficiently through standardized mass casualty triage protocols.

4. Mass Casualty Triage Algorithms: START and JumpSTART RPM Frameworks

Mass Casualty Incidents (MCI) in austere or disaster environments severely strain operational capacity, creating an immediate imbalance between medical demand and available clinical resources. Tactical disaster triage reallocates limited care to maximize survival across the greatest number of casualties. Triage is a dynamic, continuous process; initial classifications must be rapidly performed using standardized, objective algorithmic filters: the **START** (Simple Triage and Rapid Treatment) framework for adults and the **JumpSTART** adaptation for pediatric casualties.



```
                                                                Cmds
Cmds
                                                                  |         |
                                                                [RED]
[YELLOW]
```

4.1 START Triage Algorithm (Adult RPM Framework)

The START protocol evaluates adult casualties using three sequential physiological parameters: **Respirations** , **Perfusion** , and **Mental Status** (RPM).

Step 1: Initial Sorting

Direct all casualties who can hear commands and move independently to walk to a designated safe collection point. These casualties are designated **Priority 3 (Minor / Green)** .

Step 2: Respirations (R)

Evaluate non-ambulatory casualties:

- *Apneic:* Open the airway manually (head-tilt/chin-lift or jaw-thrust). If spontaneous breathing remains absent after opening the airway, classify the casualty as **Priority 0 (Expectant / Black)** . If spontaneous breathing resumes, classify as **Priority 1 (Immediate / Red)** .
- *Respiratory Rate $> 30\text{ breaths/min}$:* Classify immediately as **Priority 1 (Immediate / Red)** .
- *Respiratory Rate $10\text{--}30\text{ breaths/min}$:* Proceed to evaluate Perfusion.

Step 3: Perfusion (P)

Assess capillary refill time or radial pulse quality:

- *Capillary Refill $> 2\text{ seconds}$ or Radial Pulse Absent:* Control overt capillary/arterial bleeding immediately; classify as **Priority 1 (Immediate / Red)** .
- *Capillary Refill $\le 2\text{ seconds}$ or Radial Pulse Present:* Proceed to evaluate Mental Status.

Step 4: Mental Status (M)

Assess ability to follow simple verbal commands:

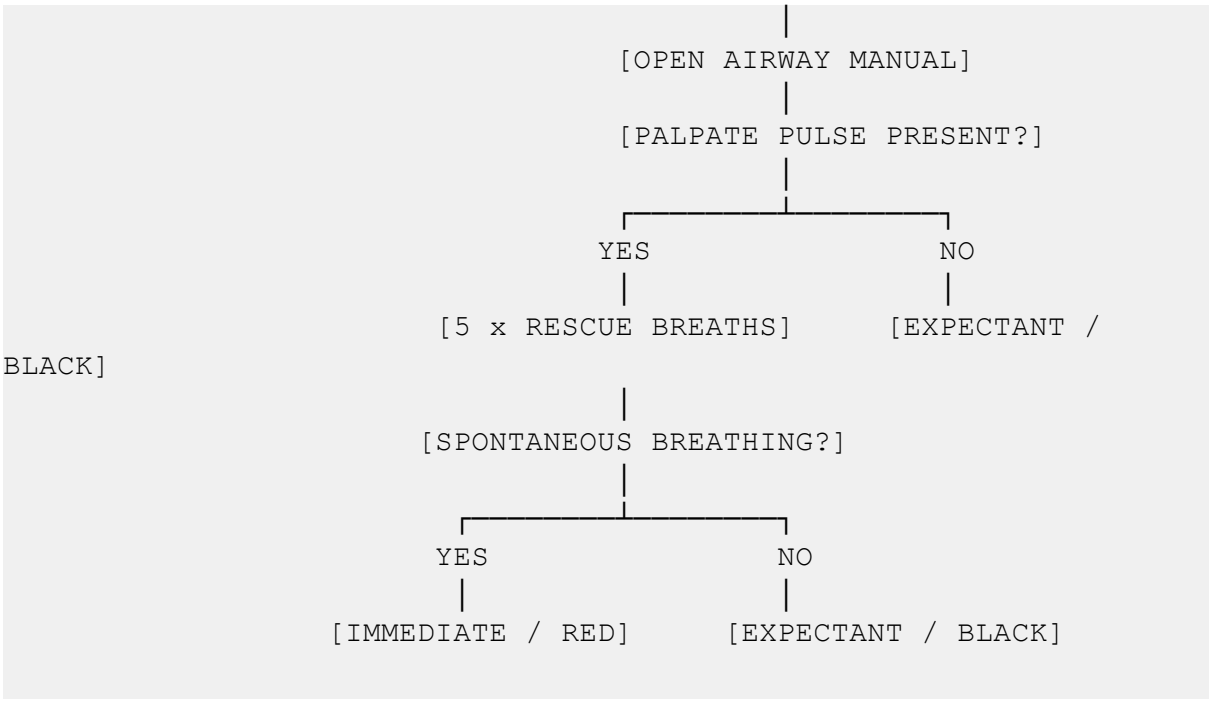
- *Unable to Follow Commands (Altered / Unconscious):* Classify as **Priority 1 (Immediate / Red)** .
- *Follows Simple Commands:* Classify as **Priority 2 (Delayed / Yellow)** .

4.2 JumpSTART Triage Algorithm (Pediatric RPM Adaptation)

Pediatric casualties possess distinct physiological responses to trauma. Hypoxia—rather than primary arterial hemorrhage—is the predominant cause of pediatric respiratory arrest and subsequent cardiac standstill. JumpSTART adapts the adult framework for children (typically identified as appearing under 8 years of age or weighing under 30 kg).

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[JUMPSTART PEDIATRIC RESCUE BREATH MODIFICATION]

                              [PEDIATRIC CASUALTY APNEIC]
```



Key Pediatric RPM Adaptations

- **The 5 Rescue Breaths Rule:** If a pediatric casualty is apneic upon opening the airway, *do not immediately classify as Expectant* . Check for a peripheral or central pulse. If a pulse is present, deliver **5 field rescue breaths** via barrier device or BVM. If spontaneous breathing resumes following rescue breathing, classify as **Priority 1 (Immediate / Red)** . If respiratory arrest persists, classify as **Priority 0 (Expectant / Black)** .
- **Pediatric Respiratory Parameters:** Normal pediatric breathing rates are faster. Respiratory rates $< 15\text{ or } > 45\text{ breaths/min}$ dictate an immediate classification of **Priority 1 (Immediate / Red)** . Rates between $15\text{ and }45\text{ breaths/min}$ proceed to perfusion evaluation.
- **AVPU Neurological Assessment:** Replace complex command checks with the AVPU scale (Alert, Verbal, Pain, Unresponsive). Casualties exhibiting Alert, verbal responsiveness, or localized pain response are classified as **Priority 2 (Delayed / Yellow)** . Casualties exhibiting non-purposeful pain posturing (decorticate/decerebrate) or complete Unresponsiveness are classified as **Priority 1 (Immediate / Red)** .

4.3 Comparative Triage Summary Table

Operational Parameter,START Algorithm (Adults),JumpSTART Algorithm (Pediatrics),Triage Action Trigger
Target Population,Adults and older adolescents.,Infants and children (typically $<8\text{ years}$ or $<30\text{ kg}$).,Initial physical triage sorting on scene.
Respiratory Thresholds,$>30\text{ bpm} \rightarrow \mathbf{Red}$; Apneic post-opening $\rightarrow \mathbf{Black}$.,$<15\text{ or }>45\text{ bpm} \rightarrow \mathbf{Red}$.,Visual breath count over 15 seconds.
Apnea Action Step,Manual airway opening only.,Manual airway opening $+$ 5 rescue breaths if pulse is present.,Immediate manual airway repositioning / rescue ventilation.

Perfusion Metrics,Capillary refill $>2\text{ s}$ or absent radial pulse $\rightarrow \mathbf{Red}$.,Absent peripheral pulses or refill $>2\text{ s} \rightarrow \mathbf{Red}$.,Radial/brachial pulse palpation or capillary press.
Neurological Check,Simple command execution check.,AVPU scale (Inappropriate/Posturing/Unresponsive $\rightarrow \mathbf{Red}$).,Verbal instruction or painful sternal rub.
Primary Triage Focus,Rapid identification of massive hemorrhage/shock.,Prevention of hypoxic cardiac arrest via respiratory rescue.,Allocation of definitive treatment priority.
Managing mass casualty scenarios efficiently highlights the critical need to preserve medical supplies, conserve pharmaceuticals, and implement expired drug scavenging protocols during extended supply chain disruptions.

5. Scavenging Expired Pharmaceuticals, SLEP Standards, and Antibiotic Stability

In prolonged disaster scenarios or isolated operational deployments, supply chains degrade, isolating medical personnel from standard re-supply. When operational stocks deplete, providers face severe ethical and clinical challenges regarding expired pharmaceutical stocks. Discarding effective medications during active drug shortages can jeopardize casualty survival; conversely, administering degraded compounds introduces risks of therapeutic failure or toxicity. Understanding empirical stability data and official shelf-life extension criteria enables safe drug scavenging when no primary alternatives exist.

```
[DRUG DOSAGE FORM STABILITY TIERING]
│
├── SOLID DOSAGE FORMS (Tablets, Capsules, Pure Powders) ──▶ EXCELLENT STABILITY
│   Retain >90% active potency for years/decades past labeled expiry if dry/unopened.
│
├── LIQUID SOLUTIONS / SUSPENSIONS / SYRUPS ──────────────▶ POOR STABILITY
│   Rapid chemical hydrolysis, precipitation, bacterial growth risk. Short post-open life.
│
└── RECONSTITUTED INJECTABLES / OPHTHALMICS ──────────────▶ EXTREME DEGRADATION RISK
    Do not use if discoloured, cloudy, or displaying visible crystal precipitation.
```

5.1 FDA Shelf-Life Extension Program (SLEP) & Stability Realities

The US Department of Defense (DoD) established the Shelf-Life Extension Program (SLEP), administered via the FDA, to evaluate the actual longevity of stockpiled pharmaceuticals and prevent unnecessary replacement costs.

Empirical SLEP Data

Analysis of thousands of pharmaceutical lots stored in unopened, original manufacturer containers under controlled conditions demonstrated that approximately 88% of tested lots (spanning 122 distinct drug products) maintained therapeutic stability past their labeled expiration dates. The average extension period granted was 66 months ($\sim 5.5\text{

years}$). Crucially, *no tested pharmaceutical lot failed stability testing within 1 year post-expiration* , and approximately 12% of evaluated lots maintained stability for over 4 years beyond their original date.

Degradation Mechanisms and Dosage Form Dynamics

- **Solid Dosage Forms:** Dry oral tablets, hard gel capsules, and sealed crystalline powders exhibit long-term stability. When protected from light and moisture, pure drug substances (e.g., ampicillin, ciprofloxacin, doxycycline) retain high potency for decades past manufacture.
- **Liquid Formulations and Suspensions:** Aqueous solutions, liquid suspensions, oral syrups, and eye drops degrade significantly faster. Water molecules drive chemical hydrolysis, breaking down core molecular structures (such as the beta-lactam ring).
- **Physical Indicators of Expiration:** Any liquid or injectable formulation exhibiting visual discoloration, cloudiness, phase separation, or crystalline precipitation must be considered degraded and discarded immediately.
- **Environmental Accelerants:** High atmospheric humidity introduces ambient water molecules, accelerating molecular breakdown. High ambient temperatures further accelerate thermal degradation. However, specific dry formulations retain efficacy even after severe environmental exposure; for instance, dry cefoxitin sodium powder subjected to severe stress conditions (40°C and 75% relative humidity) retained bactericidal efficacy for over 1.5 years.

5.2 Antibiotic Stability and Safety Profile Analysis

Solid Antibiotic Stability

Solid-form antibiotics demonstrate high stability profiles post-expiration. Long-term laboratory assays confirm that dry ciprofloxacin tablets retain 100% active drug substance up to 55 months past expiration, while pure ciprofloxacin substance retains potency for up to 300 months (25 years). Doxycycline capsules and dry powders maintain 100% potency at 27 to 50 months, while solid amoxicillin tablets show 100% active substance retention at 23 months post-expiration.

Pediatric Liquid Contraindications

Expired pediatric oral syrups and reconstituted suspensions are strictly contraindicated. Studies evaluating expired liquid pediatric syrups demonstrate significant loss of active drug potency, leading directly to clinical treatment failure and sub-therapeutic dosing. Sub-therapeutic antibiotic concentrations accelerate the selection and emergence of Multidrug-Resistant Organisms (MDROs). Reconstituted oral liquids possess short functional lifespans (typically 2 weeks to 2 months post-opening) and harbor elevated risks of bacterial contamination if un-refrigerated.

The Tetracycline Toxicity Myth vs. Modern Risk Profiles

A historical clinical concern regarding expired antibiotics involves expired tetracycline causing reversible renal tubular damage (Fanconi syndrome), characterized by nausea, vomiting, proteinuria, and metabolic acidosis 2 to 8 days post-ingestion. This toxicity was tied to specific degradation products—namely epi-anhydrotetracycline and

anhydrotetracycline—formed in older, obsolete tetracycline formulations. Modern manufacturing standards, altered excipients, and refined chemical buffering have eliminated this hazard. *There are no modern reported cases of Fanconi syndrome or acute renal toxicity associated with expired oral tetracyclines or modern derivatives such as doxycycline.*

5.3 Pharmaceutical Stability Reference Table

The following empirical data synthesizes long-term stability testing across essential antimicrobial and first-aid resuscitation compounds (Davido et al., 2024):| Antibiotic / Agent (INN) | Dosage Form | Tested Extension Period (Months) | Retained Efficacy (% Active Substance) | Mean Time Period (Months) to Active Substance Degradation || ------ | ------ | ------ | ------ | ------ || **Amoxicillin** | Tablet (21 lots) | 23 | 100% | Not degraded within tested window. || **Ampicillin** | Capsule (5 lots) | 49 | 100% | Not degraded within tested window. || **Ampicillin** | Injection-Solution (8 lots) | 57 | 87.5% | Mean degradation observed at 42 months. || **Cefazolin** | Powder (10 lots) | 82 | 80.0% | Mean degradation observed at 82 months. || **Cefoxitin** | Powder (10 lots) | 24 | 60.0% | Mean degradation observed at 24 months. || **Ceftriaxone** | Powder (4 lots) | 60 | 100% | Not degraded within tested window. || **Ciprofloxacin** | Tablet (242 lots) | 55 | 100% | Exceptional stability; 0 lot failures. || **Ciprofloxacin** | Suspension (7 lots) | 32 | 100% | Inspect for clumping or precipitation. || **Clindamycin** | Injection-Solution (31 lots) | 44 | 80.6% | Mean degradation observed at 44 months. || **Doxycycline** | Capsule (13 lots) | 50 | 100% | High stability; no modern Fanconi risk. || **Doxycycline** | Powder (31 lots) / Tablet (169 lots) | 27 | $98.2\text{--}100\%$ | Stable solid form; maintain dry storage. || **Penicillin G Benzathine** | Liquid Suspension (4 lots) | 70 | 100% | Sealed suspension maintained full potency. || **Penicillin G** | Dry Powder (15 lots) | 49 | 93.3% | High stability in unopened dry vials. || **Penicillin G** | Dry Powder (7 lots) | 70 | 28.6% | Severe degradation observed at 70 months. || **Tetracycline** | Capsule (11 lots) | 50 | 63.6% | Mean degradation observed at 32 months. || **Albuterol (Salbutamol)** | Inhaler / Solution | 240–360 | 98.0% | Retains active drug $20\text{--}30\text{ years}$ past expiry. || **Aspirin** | Solid Tablet | Up to 480 | $>90.0\%$ | Retains active substance up to 40 years if dry. || **Epinephrine** | Autoinjector | ≥ 36 | $>90.0\%$ | Autoinjectors maintain potency for at least 3 years past expiry. || **Naloxone** | Solution / Injectable | ≥ 19 | $>90.0\%$ | Retains active drug for emergency reversal past labeled date. |

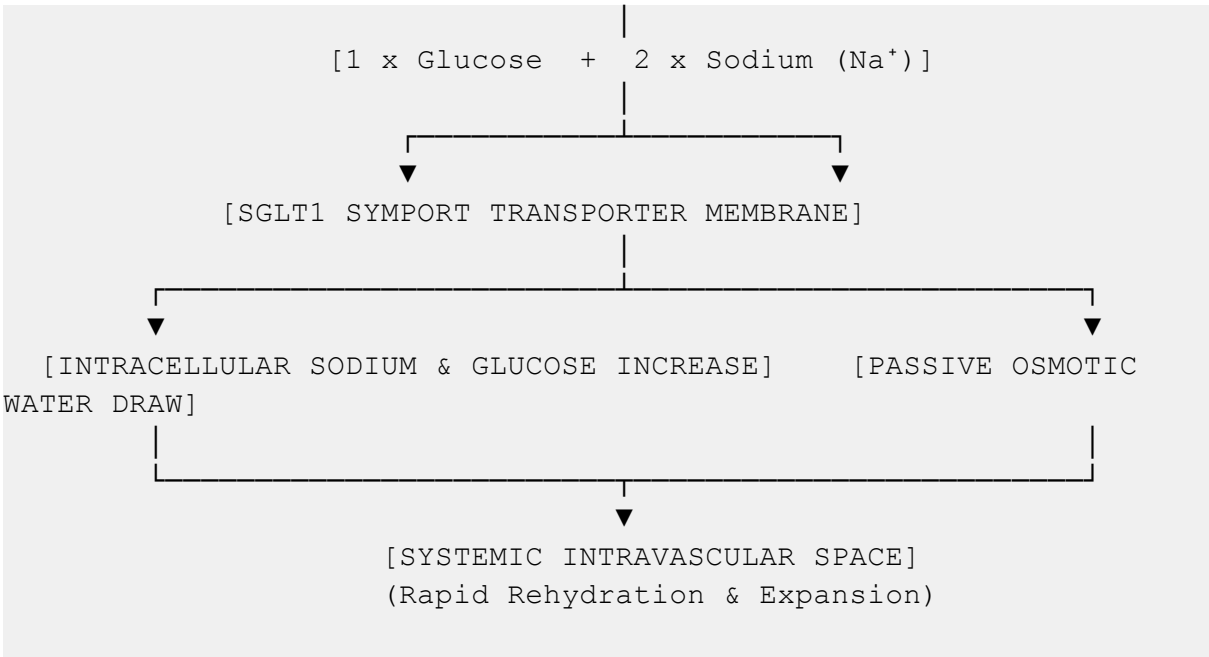
Maintaining medication efficacy supports systemic resuscitation, particularly when managing internal liquid volume shifts using oral and enteral absorption pathways.

6. Improvised Hydration, Enteral Resuscitation, SGLT1 Dynamics, and Proctoclysis

When intravenous crystalloid fluids and blood products are depleted during prolonged isolated care, systemic hypovolemia and dehydration threaten casualty survival. In austere environments where sterile parenteral access cannot be maintained, the gastrointestinal tract serves as an efficient physiological alternative for fluid resuscitation. Utilizing enteral hydration pathways bypasses the need for sterile IV infrastructure, drawing on intestinal transport mechanisms to restore circulating intravascular volume.

```
[INTESTINAL SGLT1 COTRANSPORT MECHANISM]

                          INTESTINAL LUMEN (GUT)
```



6.1 Physiology of SGLT1 Cotransport & Oral Rehydration

SGLT1 Symport Mechanics

Enteral fluid absorption relies on the **Sodium-Glucose Cotransporter 1 (SGLT1)** protein expressed on the apical membrane of intestinal enterocytes. SGLT1 operates as an active symport system, coupling the transport of two sodium ions (Na^+) alongside one molecule of glucose across the mucosal epithelium into the cell. This active movement creates an intracellular osmotic gradient, drawing water passively from the gut lumen through transcellular aquaporins into the intravascular space.

Osmolar Balance and the Hazards of Hyperosmolarity

For SGLT1 cotransport to function efficiently, enteral rehydration solutions must maintain strict **isotonic or slightly hypotonic osmolarity** (target $\sim 245\text{ mOsm/L}$). Delivering overly concentrated, hyperosmolar solutions—such as commercial undiluted sports drinks, fruit juices, or dense salt/sugar mixes—introduces high solute concentrations into the intestinal lumen.When luminal osmolarity exceeds intravascular osmolarity, water is pulled *out* of systemic circulation back into the bowel via osmosis. This acute fluid shift triggers severe osmotic diarrhea, rapid fluid loss, abdominal cramping, and accelerated hypovolemic collapse. Enteral solutions must strictly balance sodium and glucose concentrations.

6.2 WHO Oral Rehydration Salts (ORS) Standard & Improvised Formulations

Standard WHO Low-Osmolarity ORS Recipe

The standard World Health Organization (WHO) low-osmolarity formulation provides optimized solute concentrations per 1 Litre of clean water:

```
[STANDARD WHO ORS FORMULATION - PER 1 LITRE WATER]
├── Sodium Chloride (NaCl): ..................... 2.6 g
├── Glucose (Anhydrous Dextrose): ............... 13.5 g
├── Potassium Chloride (KCl): ................... 1.5 g
```

```
└── Trisodium Citrate Dihydrate: ..... 2.9 g
    (Target Osmolarity: ~245 mOsm/L)
```

Field-Improvised Rehydration Formulations

When standardized ORS packets are unavailable, prepare an emergency rehydration solution using household items dissolved in 1 Litre of boiled or purified water:

- **Level Sodium Chloride (Table Salt):** Half a level teaspoon ($\sim 3.0\text{ g}$).
- **Granulated Sugar (Sucrose) or Honey:** Six level teaspoons ($\sim 25.2\text{ g}$).
- *Safety Directive:* Stir completely until all solids dissolve fully. Taste the solution prior to administration; it should taste no saltier than human tears. *Never double the salt or sugar concentrations* , as hyperosmolar solutions induce life-threatening osmotic fluid losses.

6.3 Proctoclysis (The Murphy Drip Protocol)

Proctoclysis—the slow instillation of fluid into the rectum—provides an effective enteral hydration pathway when casualties are unconscious, experiencing persistent emesis, suffering from severe maxillofacial trauma, or unable to swallow.

```
[PROCTOCLYSIS / MURPHY DRIP SETUP]
│
├── ANATOMICAL ROUTE ──► Middle/Inferior Rectal Veins ──► Direct
Systemic Circulation
│                          (Bypasses First-Pass Hepatic Metabolism &
Upper GI Tract)
│
├── TUBE INSERTION ─────► Flexible Catheter / Nasogastric Tube
inserted 10–15 cm into rectum
│
├── FLUID SELECTION ────► Isotonic WHO ORS or 0.9% Normal Saline
(NO HYPEROSMOLAR FLUIDS)
│
└── DRIP RATE ──────────► Titrated at 200–400 mL/hr (Max ~500 mL/hr
to prevent retention leakage)
```

Anatomical Rationale

Fluids infused into the lower rectum are absorbed through the middle and inferior rectal veins. This venous network drains directly into the inferior vena cava, bypassing upper gastrointestinal disturbances, avoiding first-pass hepatic metabolism, and providing rapid systemic intravascular volume expansion without requiring sterile vascular access.

Murphy Drip Operational Setup

1. **Patient Positioning:** Place the casualty in the Sim's position (lateral decubitus with the upper leg flexed).
2. **Catheter Insertion:** Lubricate a flexible $14\text{--}18\text{ Fr}$ nasogastric or urinary catheter. Gently insert the catheter through the anal sphincter to a depth of

$10\text{--}15\text{ cm}$ into the rectal vault. Secure the catheter firmly to the gluteal fold with tape.

3. **Fluid Selection and Mucosal Safety Boundary:** Use standard WHO ORS solution or isotonic 0.9% Normal Saline. *Plain hypotonic water or concentrated hyperosmolar fluids are strictly forbidden for rectal administration.* Instilling plain water or hyperosmolar solutions causes acute mucosal edema, epithelial sloughing, severe localized pain, and rapid, unmanaged electrolyte dysregulation.
4. **Administration Rate:** Connect the catheter to a standard IV gravity drip tubing set. Regulate flow to a slow, continuous rate of $200\text{--}400\text{ mL/hour}$ (maximum 500 mL/hour). Slow rates prevent rectal distension, minimizing the defecation reflex and allowing complete fluid retention and mucosal absorption.Stabilizing intravascular volume supports peripheral perfusion, facilitating advanced local wound care and tissue maintenance.

7. Improvised Wound Management, Dakin's Solution, Delayed Primary Closure, and Biological Bio-Hacks

Combat injuries, blast wounds, and high-energy traumatic lacerations in austere environments are inherently contaminated with environmental debris, soil, and bacterial pathogens. Unsterile field conditions and delayed evacuation significantly elevate risks for acute wound infections, soft tissue necrosis, and gas gangrene. Austere wound management prioritizes rigorous infection prevention over rapid aesthetic closure, drawing on controlled debridement, delayed closure protocols, chemical antiseptics, and biological therapies.

7.1 Delayed Primary Closure (DPC) & Debridement Directives

Absolute Contraindication of Immediate Primary Closure

Immediate primary suture closure of high-energy traumatic, blast, or penetrating combat wounds in austere field environments is strictly contraindicated. Primary closure traps anaerobic spore-forming bacteria (e.g., *Clostridium perfringens*) deep within devitalized tissue pockets. The resulting anaerobic environment accelerates the development of fatal gas gangrene, severe necrotizing soft tissue infections, and systemic sepsis.

```
[DELAYED PRIMARY CLOSURE (DPC) TIMELINE & INSPECTION CRITERIA]
│
├── DAY 0: INITIAL PRESENTATION & DEBRIDEMENT
│   ├── Action: Surgical excision of non-viable tissue using "4 Cs" + 3-9 L irrigation.
│   └── Dressing: Pack loosely with wet-to-dry saline/Dakin's gauze. LEAVE WOUND OPEN.
│
├── DAYS 1 TO 3: SERIAL RE-INSPECTION (DAILY DRESSING CHANGES)
│   ├── Exudate Assessment: Monitor for serosanguinous vs. thick purulent drainage.
│   ├── Margin Evaluation: Inspect for advancing erythema, warmth, or skin edge necrosis.
│   ├── Odor & Undermining: Check for foul anaerobic odor or deep tissue pocketing.
```

```
│    └── Debridement: Perform selective re-excision of newly
demarcated necrotic tissue.
│
└── DAYS 4 TO 5: DEFINITIVE DELAYED PRIMARY CLOSURE
    ├── Criteria: Healthy beefy-red granulation tissue (100% bed
coverage), zero purulence.
    └── Execution: Approximate edges loosely with monofilament
sutures or surgical staples.
```

DPC Step-by-Step Field Protocol

1. **Initial Debridement (Day 0):** Surgical excision of all devitalized, non-viable, or heavily contaminated tissue. Assess tissue viability using the "4 Cs": **Color** (healthy pink vs. pale/black), **Consistency** (firm vs. friable), **Contractility** (muscular response to pinch), and **Capillary Bleeding** (active microvascular bleeding upon incising).
2. **Low-Pressure Irrigation:** Irrigate the open wound bed thoroughly using large volumes (3–9 Litres) of clean, potable water or sterile saline. *Avoid high-pressure pulse lavage* , which forces bacteria and microscopic debris deeper into uninjured tissue planes.
3. **Loose Packing:** Pack the open wound bed loosely with sterile gauze moistened with saline or full/half-strength Dakin's solution. Do not pack tightly, as excessive pressure restricts microvascular perfusion. Cover with an outer protective dressing.
4. **Serial Re-Evaluation (Days 1–3):** Perform daily dressing changes under clean conditions. Inspect exudate color (clear/serosanguinous vs. yellow/green purulent), evaluate tissue margins for advancing erythema or edge undermining, and smell for foul anaerobic odor. Perform secondary surgical debridement as needed.
5. **Definitive Delayed Closure (Days 4–5):** If the wound bed demonstrates healthy, beefy-red granulation tissue covering 100% of the surface without signs of localized or systemic infection at 4 to 5 days, perform Delayed Primary Closure. Approximate wound edges loosely using simple interrupted monofilament sutures, surgical staples, or skin tapes.

7.2 Improvised Antiseptics: Dakin's Solution Preparation

Dakin's solution (Sodium Hypochlorite) is a potent, broad-spectrum topical antiseptic effective against Gram-positive/Gram-negative bacteria, fungi, and spore-forming organisms. It can be prepared in field settings using household liquid bleach (5.25% sodium hypochlorite solution).

```
[DAKIN'S SOLUTION FIELD PREPARATION - 1 LITRE WATER]
│
├── STEP 1: Boil 1 Litre of clean water for 15 minutes; allow to
cool.
├── STEP 2: Add Sodium Bicarbonate (Baking Soda) buffer: 1/2
teaspoon (~2.5 g).
└── STEP 3: Add Liquid Household Bleach (5.25% Sodium
Hypochlorite):
    │
```

```
    ├── FULL STRENGTH (0.50%): ... Add 95 mL Bleach (~3.2 oz /
~1/3 cup)
    ├── HALF STRENGTH (0.25%): ... Add 48 mL Bleach (~1.6 oz / ~3
tablespoons)
    └── QUARTER STRENGTH (0.125%): Add 24 mL Bleach (~0.8 oz /
~1.5 tablespoons)
```

Preparation Directives and Shelf Life

- *Buffering:* Adding sodium bicarbonate (baking soda) buffers the solution's pH, minimizing localized chemical tissue irritation and pain during irrigation.
- *Storage and Degradation:* Store Dakin's solution in dark, light-resistant glass or opaque plastic containers. Sodium hypochlorite degrades rapidly when exposed to heat and direct sunlight. Unbuffered solutions must be remade daily; properly buffered solutions stored in light-proof containers maintain stability for up to 7 days.

7.3 Biological Bio-Hacks and Tool Sterilization

Medical-Grade and Raw Honey

Unprocessed honey functions as a biological topical antimicrobial agent. Honey contains high sugar concentrations that exert a strong osmotic draw, pulling water out of bacterial cells and inhibiting pathogen replication. Furthermore, honey naturally generates low levels of hydrogen peroxide via enzymatic glucose oxidase activity while creating an acidic wound interface (pH 3.2–4.5) that promotes tissue healing and inhibits bacterial colonization.

Sphagnum Moss

Clean, dried Sphagnum moss can serve as a field-expedient absorbent wound dressing. Sphagnum moss holds high volumes of fluid and creates an inherently acidic localized microenvironment that inhibits bacterial proliferation.

Maggot Debridement Therapy (MDT)

In cases of deep, necrotic wounds where surgical instruments are limited, larvae of the green bottle fly (*Lucilia sericata*) can perform targeted biological debridement. These larvae secrete proteolytic enzymes that selectively liquefy devitalized, necrotic tissue while sparing healthy granulation tissue. The larvae ingest the liquefied dead tissue and secrete ammonia, alkalinizing the wound bed and suppressing bacterial growth.

Field Equipment Sterilization Protocols

When standard autoclaves are unavailable, sterilize surgical instruments using field-expedient methods:

- **Pressure Cooking:** Place instruments in a standard pressure cooker at 121°C (15 psi) for at least 30 minutes. This is the most reliable field method for destroying heat-resistant bacterial endospores.
- **Boiling Water:** Submerge instruments completely in boiling water (100°C) for a minimum of 30 minutes. *Note: Boiling destroys vegetative bacteria but does not guarantee destruction of all bacterial endospores.*

- **Chemical Immersion:** Submerge instruments in full-strength Dakin's solution (0.5%) or 70% isopropyl alcohol for at least 30 minutes, followed by rinsing with sterile saline.
- **Flame Sterilization:** Direct exposure of metal instrument tips to an open flame until red-hot. Allow instruments to air-cool completely on a sterile surface prior to tissue contact.Controlling local tissue trauma and executing surgical debridement requires managing casualty pain and systemic physiological stress.

8. Austere Pain Management, Triple-Option Analgesia, Ketamine Protocols, and Opioid Reversal

Providing immediate pain control in tactical and prolonged field care settings prevents traumatic shock, blunts harmful sympathetic surge, and mitigates long-term psychological trauma. However, battlefield analgesia must carefully balance effective pain relief against the risks of respiratory depression, loss of protective airway reflexes, and acute hemodynamic collapse in hypotensive casualties.

```
[TCCC TRIPLE-OPTION ANALGESIA FRAMEWORK]
│
├── OPTION 1: MILD TO MODERATE PAIN (Casualty Able to Fight / Conscious)
│    └── Combat Wound Medication Pack (CWMP): Acetaminophen + Meloxicam
│
├── OPTION 2: MODERATE TO SEVERE PAIN (Casualty NOT in Shock / Respiratory Distress)
│    └── Oral Transmucosal Fentanyl Citrate (OTFC): 800 mcg transmucosally
│
└── OPTION 3: MODERATE TO SEVERE PAIN (Casualty IN SHOCK or At Risk of Shock)
     └── Ketamine IV/IO/IM Protocols (Preserves spontaneous respiration & blood pressure)
```

8.1 TCCC Triple-Option Analgesia Framework

TCCC dictates a three-tiered analgesia framework based on injury severity, hemodynamic stability, and operational status:

Option 1: Mild to Moderate Pain (Casualty Maintains Combat Capability)

Administer the **Combat Wound Medication Pack (CWMP)** :

- *Acetaminophen:* 1000 mg orally every 8 hours.
- *Meloxicam:* 15 mg orally once daily.
- *Clinical Directive:* Non-sedating oral agents control pain without altering mental status, allowing the casualty to retain fighting capability and cognitive awareness.

Option 2: Moderate to Severe Pain (Casualty NOT in Shock or Respiratory Distress)

Administer **Oral Transmucosal Fentanyl Citrate (OTFC)** :

- *Dose:* 800 mcg transmucosally.

- *Administration:* Place the fentanyl lozenge between the cheek and lower gum; instruct the casualty to suck on the lozenge rather than chew or swallow it.
- *Clinical Directive:* Do not administer fentanyl to casualties exhibiting signs of hypovolemic shock, severe hypotension ($\text{SBP} < 90\text{ mmHg}$), or respiratory depression. Secure the lozenge to the casualty's finger with tape to allow it to drop away if the casualty becomes over-sedated.

Option 3: Moderate to Severe Pain (Casualty IN SHOCK or At Risk of Shock)

Administer **Ketamine** :

- Ketamine provides potent analgesia and sedation while maintaining intrinsic sympathetic vascular tone, spontaneous respiratory drive, and airway protective reflexes. It is the drug of choice for casualties exhibiting hypovolemic or traumatic shock.

8.2 Ketamine Dosing Protocols and Casualty Safety

Dosing Specifics

- **Sub-Dissociative Analgesia Dosing:** $20\text{--}30\text{ mg}$ IV/IO slow push (over 1 minute), repeated every 20 minutes as needed; or $50\text{--}100\text{ mg}$ IM/IN, repeated every $30\text{--}40\text{ minutes}$.
- **Procedural Sedation Dosing:** $1\text{--}2\text{ mg/kg}$ IV/IO push for advanced airway placement or surgical procedures.
- **Prolonged IM Sedation Dosing:** $3\text{--}4\text{ mg/kg}$ IM for extended field sedation when vascular access is absent.
- **Continuous IV Drip:** Titrated ketamine infusion to maintain long-term procedural analgesia and sedation.

```
                  [MANDATORY CASUALTY DISARMING DIRECTIVE]
┌──────────────────────────────────────────────────────────────────────────
────────┐
│ WARNING: Casualties receiving sedating analgesics (Ketamine,
OTFC,     │
│ Opioids, Benzodiazepines) can experience vivid emergence
delirium,     │
│ hallucinations, or altered perception.
│
│
│
│ MANDATORY ACTION: IMMEDIATELY DISARM the casualty of all primary
and    │
│ secondary weapons, sidearms, grenades, knives, and explosive
ordnance    │
│ PRIOR TO ADMINISTERING sedating analgesic agents.
│
└──────────────────────────────────────────────────────────────────────────
────────┘
```

Hemodynamic Monitoring

Monitor blood pressure and pulse rate closely—every $1\text{--}2\text{ minutes}$ during procedural administration, and every $3\text{--}5\text{ minutes}$ for at least 15 minutes post-administration. Ketamine maintains blood pressure in most casualties; however, in severely depleted, un-resuscitated casualties with exhausted end-stage sympathetic reserves, ketamine can precipitate transient hypotension. Resuscitate with blood products or crystalloid as indicated.

8.3 Opioid Toxicity and Titrated Naloxone Reversal

Opioid Overdose Symptoms

Opioid administration (fentanyl, morphine, hydromorphone) carries risks of accidental overdose, presenting with respiratory depression (respiratory rate $<8\text{ breaths/min}$ or apnea), profound central nervous system depression, hypoxemia, and bilateral miosis (pinpoint pupils).

Titrated Naloxone Administration

When administering Naloxone (Narcan) for opioid-induced respiratory depression, *do not administer a rapid, large IV bolus (2.0 mg)* unless the casualty is in cardiac arrest. Sudden complete opioid reversal instantly triggers acute opioid withdrawal syndrome, violent agitation, massive sympathetic surge, severe pain crisis, and acute pulmonary edema.

```
[TITRATED NALOXONE REVERSAL PROTOCOL]
│
├── DILUTION: Dilute 0.4 mg Naloxone ampoule in 9 mL Normal Saline
(Total = 10 mL)
│             (Yields Concentration of 40 mcg/mL)
│
├── DOSING:   Administer 40 mcg (1 mL) IV/IO every 1–2 minutes
│
└── TARGET:   Titrate strictly until spontaneous respiratory drive
resumes (RR 10–12 bpm)
              STOP administration once respirations stabilize; DO
NOT fully reverse pain control.
```

Half-Life Mismatch Warning and Re-Monitoring

Operators must account for the critical half-life mismatch between naloxone and synthetic opioids. Naloxone possesses a short terminal half-life of approximately $30\text{--}90\text{ minutes}$, whereas longer-acting or transmucosal opioids (such as OTFC lozenges, morphine, or hydromorphone) exhibit prolonged elimination half-lives lasting 2 to 6 hours. As naloxone clears from central opioid receptors, the casualty is at high risk for re-narcotization and recurrent, fatal respiratory arrest. Continuously monitor respiratory rate and saturation for at least $2\text{--}4\text{ hours}$ post-reversal, preparing to administer repeated titrated doses of naloxone as required.Pain management stabilizes the casualty sufficiently to allow localized orthopedic immobilization and surgical decompression.

9. Tactical Field Splinting, Compartment Syndrome Recognition (6 Ps), and Surgical Decompression

High-energy combat trauma, blast exposure, and structural collapse frequently cause complex long-bone fractures and severe soft-tissue crush injuries. Unstabilized fractures drive internal blood loss, accelerate soft-tissue destruction, and significantly elevate the risk of fat embolism syndrome. Proper field immobilization protects adjacent neurovascular bundles, reduces hemorrhage, and prevents secondary tissue injury. However, progressive tissue swelling within closed fascial compartments introduces the threat of acute compartment syndrome.

9.1 Tactical Field Splinting Rules

Execute fracture field splinting adhering to five mandatory rules:

1. **Mandatory Pre- and Post-Splinting Neurovascular Checks:** Assess, document, and continuously re-evaluate distal Pulse, Motor, and Sensory (PMS) status *immediately before* applying any splint, and *re-check immediately after* completion of splinting. If PMS deteriorates post-application, loosen and readjust the splint instantly.
2. **Immobilize Joint Above and Joint Below:** To achieve true fracture stabilization, rigid splinting must immobilize both the joint immediately proximal and the joint immediately distal to the fracture site (e.g., splinting the wrist and elbow for a mid-shaft radius/ulna fracture).
3. **Traction Splinting Constraints:** Femur fractures require traction splinting to reduce muscle spasm, align bone fragments, and decrease volume loss into the thigh cavity. *Traction splints are strictly contraindicated if the casualty presents with a concurrent ipsilateral ankle injury, open knee dislocation, or lower leg/tibial fracture.*
4. **Gentle Alignment of Severe Angulation:** If an angulated extremity fracture presents with absent distal pulses, apply gentle, steady manual inline traction along the long axis of the limb to restore physical alignment and re-establish distal arterial flow. If resistance is met or severe pain is elicited, fix the limb in its position of deformity.
5. **Open Fracture Management:** Cover exposed bone ends with clean dressings. Do not force exposed, contaminated bone ends back into the deep soft-tissue envelope; irrigate exposed ends thoroughly prior to reduction attempts.

9.2 Clinical Recognition of Compartment Syndrome (The 6 Ps)

Acute compartment syndrome occurs when interstitial tissue pressure within a closed fascial compartment exceeds capillary perfusion pressure, causing muscle ischemia, tissue necrosis, and loss of limb function.

```
[THE 6 Ps OF COMPARTMENT SYNDROME]
│
├── 1. PAIN ──────────────► EARLY SIGN: Disproportionate to injury; severe with passive muscle stretch
├── 2. PARESTHESIA ──────► EARLY SIGN: Burning, tingling, or sensory loss along nerve distribution
├── 3. PALLOR ────────────► INTERMEDIATE SIGN: Pale, dusky skin tone; decreased capillary refill
```

```
├── 4. POIKILOTHERMIA ➔ INTERMEDIATE SIGN: Limb matches ambient
temperature; cool skin tone
├── 5. PARALYSIS ──────➔ LATE SIGN: Complete loss of voluntary
motor function; muscular necrosis
└── 6. PULSELESSNESS ──➔ LATE SIGN: Absent distal pulses.
                         WARNING: Waiting for pulselessness
guarantees irreversible muscle loss!
```

Diagnostic Directives

- *Early Signs:* Pain out of proportion to the apparent physical injury, coupled with severe pain elicited by passive stretching of the affected muscle groups (e.g., passive extension of the toes causing severe deep calf pain), represents the single most sensitive early physical indicator.
- *Crucial Warning:* **Do not wait for pulselessness to diagnose compartment syndrome.** Arterial pulses typically persist until intracompartmental pressures exceed mean arterial pressure. Waiting for pulselessness or complete motor paralysis leads to permanent muscle infarction, irreversible contractures, systemic toxic release, and functional limb loss.

9.3 Surgical Decompression Directives (Escharotomy vs. Fasciotomy)

When compartment pressures rise uncontrollably, surgical decompression is the sole intervention capable of preventing tissue necrosis. The provider must differentiate between full-thickness burn eschar compression and deep muscular compartment syndrome:

```
[DECOMPRESSION TECHNIQUES: ESCHAROTOMY VS FASCIOTOMY]
│
├── ESCHAROTOMY (Burn Circumferential Compression)
│   ├── Target: Incises non-viable burn ESCHAR down to
subcutaneous fat.
│   ├── Boundary: DOES NOT PENETRATE THE DEEP MUSCULAR FASCIA.
│   └── Goal: Relieves chest wall restrictions or extremity
constriction from inelastic burn eschar.
│
└── FASCIOTOMY (Deep Muscular Compartment Syndrome)
    ├── Target: Incises skin, subcutaneous fat, AND LOOSENS DEEP
MUSCLE FASCIA.
    ├── Boundary: Full surgical division of fascial envelopes
across all muscular compartments.
    └── Goal: Relieves elevated intramuscular pressure from blunt
trauma, crush, or reperfusion.
```

Escharotomy

- *Indications:* Circumferential full-thickness (3rd-degree) thermal or chemical burns involving the torso or limbs. Circumferential burn eschar forms an inelastic ring, restricting chest wall expansion (causing asphyxiation) or constricting limb circulation.
- *Execution:* Make longitudinal surgical incisions through the non-viable, leathery burn eschar down to the underlying subcutaneous fat layer. *Do not incise the deep*

muscular fascia. Incisions run along the lateral and medial mid-axial lines of the limbs or along the anterior axillary lines of the chest wall.

Fasciotomy

- *Indications:* Confirmed acute deep muscle compartment syndrome resulting from severe crush mechanisms, high-energy fractures, heavy blast contusions, or prolonged arterial occlusion post-tourniquet conversion.
- *Anatomical Landmarks and Execution Boundaries:*
- *Lower Leg (Two-Incision, Four-Compartment Fasciotomy):* Make an **anterolateral incision** parallel to the fibular shaft, placed 2 cm lateral to the anterior tibial crest, extending from below the fibular head to above the lateral malleolus. Identify the intermuscular septum separating the anterior and lateral compartments, and incise the fascia overlying both compartments longitudinally. Make a **posteromedial incision** 2 cm posterior to the posterior border of the tibia, incising the superficial posterior compartment fascia, then release the deep posterior compartment fascia off the posterior tibial bone. *Failure to release all four compartments results in muscle necrosis.*
- *Thigh (Two-Incision Fasciotomy):* Make a lateral incision from the greater trochanter to the lateral femoral condyle, incising the fascia lata and opening the anterior and posterior compartments. Make a secondary medial incision to release the adductor compartment.Unmanaged tissue compression and muscle necrosis trigger dangerous systemic metabolic complications upon extrication, leading directly to crush syndrome.

10. Crush Syndrome Pathophysiology, Reperfusion Injury Prevention, and the 4-Step Hyperkalemia Protocol

Crush syndrome is the systemic metabolic manifestation of prolonged compressive trauma to skeletal muscle tissue. Structural entrapment under heavy rubble, vehicle collapse, or blast debris isolates the damaged tissue; however, extrication and subsequent pressure release unleash a lethal cascade. The sudden restoration of blood flow into previously compressed tissue (reperfusion) washes toxic cellular contents directly into systemic circulation, transforming a localized limb injury into a life-threatening systemic disaster.

```
[CRUSH SYNDROME LETHAL METABOLIC CASCADE]
│
├── PROLONGED COMPRESSION ──► Skeletal Muscle Ischemia & Cellular Membrane Lysis
│
├── EXTRICATION (REPERFUSION)
│   ├── Systemic Release of Potassium (K⁺) ────► Lethal Cardiac Arrhythmias & Arrest
│   ├── Systemic Release of Myoglobin ──────────► Acute Tubular Necrosis & Renal Failure
│   └── Systemic Release of Lactic Acid/Purines ─► Severe Metabolic Acidosis & Shock
│
└── REPERFUSION SHOCK ─────► Plasma Shift into Damaged Muscle ──► Hypovolemic Collapse
```

10.1 Pathophysiology of Crush Injury and Reperfusion Cascade

Cellular Mechanisms

1. **Rhabdomyolysis:** Mechanical pressure drives cellular membrane rupture (lysis) across skeletal muscle cells (myocytes).
2. **Sequestration and Reperfusion Shock:** Upon extrication, systemic arterial blood flows into damaged capillary beds. Plasma shifts rapidly out of intravascular circulation into the traumatized, necrotic muscle, causing severe hypovolemic shock.
3. **Myoglobinuria and Acute Kidney Injury (AKI):** Large quantities of myoglobin are released into circulation. Filtered myoglobin precipitates within the renal tubules under acidic conditions ($\text{pH} < 5.6$), forming dense casts that obstruct the nephrons, causing acute tubular necrosis and acute renal failure.
4. **Hyperkalemic Cardiac Arrest:** Massive quantities of intracellular potassium (K^+) flood systemic circulation instantly upon pressure release. Elevated serum potassium disrupts the cardiac membrane resting potential, causing fatal ventricular fibrillation or asystole.

Pre-Extrication Management Protocol

- **Aggressive Pre-Hydration:** Establish large-bore IV/IO access and initiate intravenous crystalloid loading *prior to lifting the crushing weight off the casualty* . Administer isotonic fluids at a rate of $1.0\text{--}1.5\text{ Litres/hour}$ in adults to establish high renal clearance and dilute incoming systemic toxins.
- **Deliberate Tourniquet Isolation:** If a casualty has experienced total structural entrapment of a limb for $>2\text{ hours}$ and pre-extrication IV hydration cannot be established prior to release, *apply arterial tourniquets tightly around the compressed limbs directly prior to lifting the weight* . Tourniquet placement isolates the necrotic limb, physically preventing the systemic surge of potassium, myoglobin, and acidotic byproducts from entering central circulation. Leave tourniquets applied until the casualty can be resuscitated under direct clinical monitoring.

10.2 Emergency Hyperkalemia Mitigation Protocol

Hyperkalemia is the most immediate life-threatening metabolic emergency in crush syndrome. Manage acute hyperkalemia using a structured 4-step medical protocol:

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[EMERGENCY 4-STEP HYPERKALEMIA PROTOCOL]
│
├── STEP 1: CARDIAC MEMBRANE STABILIZATION
│   └── Calcium Gluconate (10%): 10–20 mL IV/IO over 2–5 minutes
(or Calcium Chloride 10% 5–10 mL)
│       Mechanism: Antagonizes potassium-mediated membrane
excitability. Prevents cardiac arrest.
│
├── STEP 2: INTRACELLULAR POTASSIUM SHIFTING
│   ├── Regular Insulin: 10 Units IV push
│   ├── COMBINED WITH Dextrose 50% (D50W): 50 mL (25 g) IV push
(Prevents Hypoglycemia)
│   └── Adjunct: Inhaled Albuterol (Salbutamol): 10–20 mg
nebulised
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│       Mechanism: Drives potassium from extracellular plasma back
into intracellular space.
│
├── STEP 3: ACIDOSIS CORRECTION & RENAL PROTECTION
│    └── Sodium Bicarbonate (8.4%): 50–100 mEq IV slow infusion
│       Mechanism: Alkalinises systemic pH (shifts K⁺ into cells) &
urine (prevents myoglobin cast precipitation).
│
└── STEP 4: POTASSIUM EXCRETION & BINDING
    └── Kayexalate (Sodium Polystyrene Sulfonate): 15–30 g orally
or enterally
       Mechanism: Binds potassium in the gastrointestinal tract
for definitive elimination.
```

Step 1: Cardiac Membrane Stabilization

- *Action:* Administer **Calcium Gluconate 10%** ($10\text{--}20\text{ mL}$ IV/IO over $2\text{--}5\text{ minutes}$) or **Calcium Chloride 10%** ($5\text{--}10\text{ mL}$ IV/IO).
- *Ionized Calcium and Route Boundary:* **Calcium Chloride delivers three times the concentration of elemental/ionized calcium compared to Calcium Gluconate.** Due to severe vessel irritation and the extreme risk of skin/soft tissue necrosis upon extravasation, Calcium Chloride must ideally be administered through a central line or a large-bore, verified peripheral IV line.
- *Rationale:* Calcium directly antagonizes the deleterious electrophysiological effects of elevated potassium on cardiac myocytes, restoring normal resting membrane potential and preventing immediate hyperkalemic ventricular arrhythmias. *Onset: $1\text{--}3\text{ minutes}$; Duration: $30\text{--}60\text{ minutes}$.*

Step 2: Intracellular Potassium Shifting

- *Action:* Administer **Regular Insulin** (10 Units IV push) immediately combined with **Dextrose 50% (D50W)** (50 mL 25 g IV push). Concurrently, administer high-dose **Nebulized Albuterol** ($10\text{--}20\text{ mg}$ inhaled).
- *Post-Administration Glucose Monitoring Intervals:* Insulin drives glucose into cells along with potassium; however, exogenous glucose clearance rates vary. Operators must perform mandatory point-of-care blood glucose checks at **30, 60, and 120 minutes** post-administration to detect and treat delayed, life-threatening hypoglycemia.
- *Rationale:* Insulin stimulates cell-membrane $\text{Na}^+/\text{K}^+ \text{ATPase}$ pumps, driving extracellular potassium back into intracellular spaces. Dextrose prevents severe hypoglycemia. Beta-2 agonist activity from high-dose albuterol further shifts potassium intracellularly. Stability studies confirm that albuterol salbutamol formulations retain 98% active substance for $20\text{--}30\text{ years}$, providing a reliable field option. *Onset: $10\text{--}20\text{ minutes}$; Duration: $4\text{--}6\text{ hours}$.*

Step 3: Acidosis Correction and Urinary Alkalinization

- *Action:* Administer **Sodium Bicarbonate 8.4%** ($50\text{--}100\text{ mEq}$ IV slow infusion).
- *Rationale:* Bicarbonate elevates systemic pH, inducing metabolic alkalosis that drives potassium into cells in exchange for hydrogen ions. Additionally, alkalinizing urine ($\text{target pH} > 6.5$) prevents myoglobin precipitation in renal tubules, reducing acute tubular damage. *Onset: $15\text{--}30\text{ minutes}$* .

Step 4: Potassium Excretion and Elimination

- *Action:* Administer **Sodium Polystyrene Sulfonate (Kayexalate)** ($15\text{--}30\text{ g}$ orally or enterally).
- *Rationale:* Kayexalate acts as a potassium-binding resin within the gastrointestinal tract, exchanging sodium ions for potassium ions and facilitating definitive elimination via the bowel. *Onset: $2\text{--}12\text{ hours}$* .

10.3 Comprehensive Summary Matrix for Hyperkalemia Management

Medication,Primary Mechanism of Action,Austere Dose & Route,Onset of Action,Critical Monitoring Requirements
Calcium Gluconate (10%),Membrane stabilization; antagonizes cardiac membrane excitability.,$10\text{--}20\text{ mL}$ IV/IO slow push ($2\text{--}5\text{ mins}$).,$1\text{--}3\text{ minutes}$,Continuous EKG/pulse monitoring; check for tissue extravasation necrosis.
Calcium Chloride (10%),Potent membrane stabilization; delivers $3\times$ ionized calcium concentration vs gluconate.,$5\text{--}10\text{ mL}$ IV/IO slow push ($2\text{--}5\text{ mins}$).,$1\text{--}3\text{ minutes}$,Central or large-bore peripheral line required. Extravasation causes severe necrosis.
Regular Insulin $+$ Dextrose (50%),Drives extracellular potassium into cells via $\text{Na}^+/\text{K}^+ \text{ATPase}$ stimulation.,10 Units Regular Insulin IV $+$ 50 mL D50W IV push.,$10\text{--}20\text{ minutes}$,"Mandatory blood glucose checks at 30, 60, and 120 minutes for delayed hypoglycemia."
Albuterol (Salbutamol),Beta-2 receptor agonist; drives intracellular potassium shifting.,$10\text{--}20\text{ mg}$ nebulized via mask.,$15\text{--}30\text{ minutes}$,"Monitor for tachycardia, muscular tremor, and anxiety."
Sodium Bicarbonate (8.4%),Systemic alkalinization; urine alkalinization to prevent myoglobin precipitation.,$50\text{--}100\text{ mEq}$ slow IV infusion.,$15\text{--}30\text{ minutes}$,"Monitor for tetany, hypocalcemia, and fluid overload."
Kayexalate (Polystyrene Sulfonate),Enteral potassium-binding resin; increases fecal elimination.,$15\text{--}30\text{ g}$ orally or rectally.,$2\text{--}12\text{ hours}$,"Monitor for hypokalemia, bowel obstruction, and sodium overload."
Preserving life and limb under extreme disaster constraints requires rigorous adherence to evidence-based Prolonged Field Care protocols, maintaining systematic discipline across tactical trauma, austere pharmacology, and resource-constrained resuscitation.